

AI ASSISTED CODING

ASSIGNMENT – 14.5

NAME: Varshitha. G

HTNO: 2303A51103

Task 1 – Image Gallery Website

Create an image gallery web page.

Requirements:

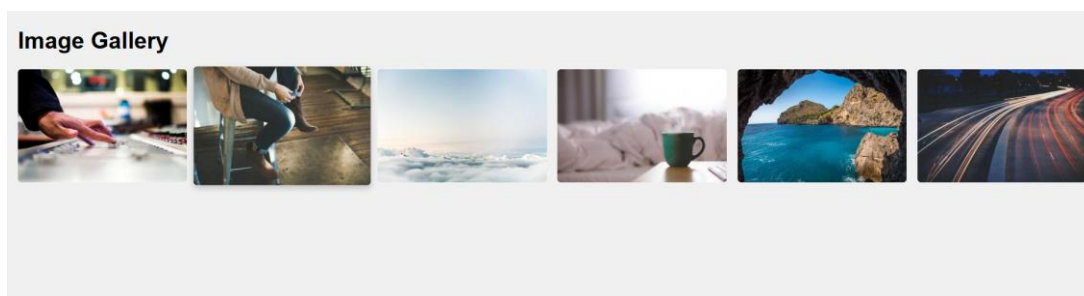
- Display images in a grid layout.
- Use AI to:
 - o Optimize layout using CSS Grid.
 - o Add hover zoom or overlay effects.
 - o Ensure mobile responsiveness.

```
<!DOCTYPE html>
<html>
<head>
  <title>Image Gallery</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f0f0;
      margin: 0;
      padding: 20px;
    }
    .gallery {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
      gap: 15px;
    }
    .gallery img {
      width: 100%;
      height: auto;
      border-radius: 5px;
      transition: transform 0.3s ease, box-shadow 0.3s ease;
    }
    .gallery img:hover {
      transform: scale(1.05);
      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    }
    @media (max-width: 600px) {
      .gallery {
        grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
      }
    }
  </style>
</head>
<body>
```

```

</style>
</head>
<body>
  <h1>Image Gallery</h1>
  <div class="gallery">
    
    
    
    
    
    
  </div>
</body>
</html>

```



Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

- Display user image, name, role, and social media links.
- Use AI to:
 - o Suggest modern UI colors and fonts.
 - o Design the card using Flexbox.
 - o Add hover animation and shadow effects.

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Profile Card</title>
5    <style>
6      body {
7        font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
8        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
9        margin: 0;
10       padding: 20px;
11       min-height: 100vh;
12       display: flex;
13       justify-content: center;
14       align-items: center;
15     }
16     .card {
17       display: flex;
18       flex-direction: column;
19       align-items: center;
20       background-color: rgba(255, 255, 255, 0.95);
21       border-radius: 15px;
22       box-shadow: 0 4px 20px rgba(0, 0, 0, 0.1);
23       padding: 30px;
24       transition: transform 0.3s ease, box-shadow 0.3s ease;
25       max-width: 300px;
26     }
27     .card:hover {
28       transform: translateY(-10px);
29       box-shadow: 0 8px 30px rgba(0, 0, 0, 0.2);
30     }
31     .card img {
32       width: 150px;
33       height: 150px;
34       border-radius: 50%;
35       object-fit: cover;
36       margin-bottom: 20px;

```

```

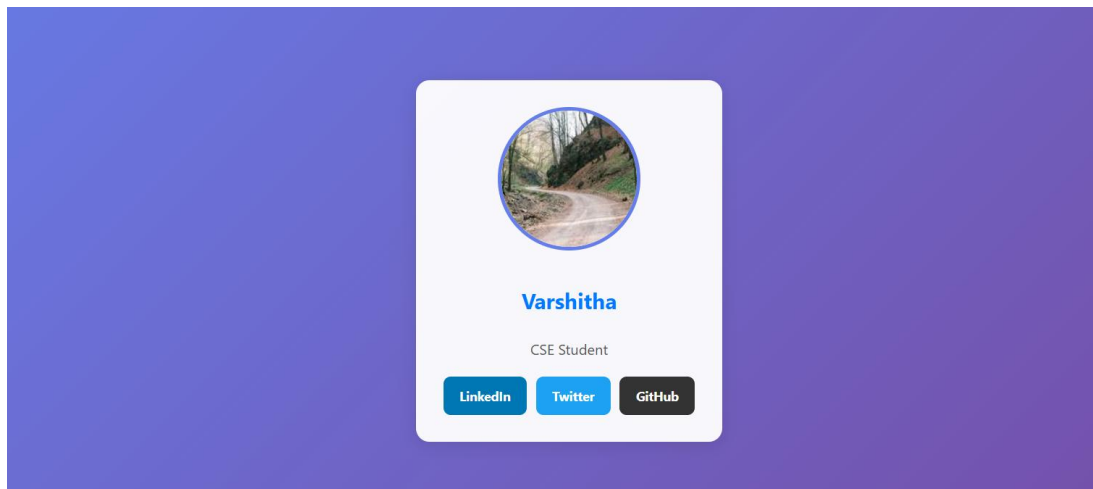
8      }
9      .card h2 {
10         color: #007bff;
11         margin-bottom: 10px;
12         font-size: 24px;
13     }
14     .card p {
15         color: #555;
16         margin-bottom: 20px;
17         font-size: 16px;
18     }
19     .social-links {
20         display: flex;
21         gap: 10px;
22     }
23     .social-links a {
24         display: inline-block;
25         padding: 12px 18px;
26         color: white;
27         text-decoration: none;
28         border-radius: 8px;
29         font-weight: bold;
30         transition: transform 0.2s, box-shadow 0.2s, background-color 0.2s;
31         font-size: 14px;
32     }
33     .social-links a:hover {
34         transform: translateY(-3px);
35         box-shadow: 0 6px 12px rgba(0, 0, 0, 0.3);
36     }
37     .linkedin {
38         background-color: #0077b5;
39     }
40     .linkedin:hover {
41         background-color: #005885;

```

```

110     }
111     .linkedin {
112         background-color: #0077b5;
113     }
114     .linkedin:hover {
115         background-color: #005885;
116     }
117     .twitter {
118         background-color: #1da1f2;
119     }
120     .twitter:hover {
121         background-color: #0d95e8;
122     }
123     .github {
124         background-color: #333;
125     }
126     .github:hover {
127         background-color: #24292e;
128     }
129     </style>
130 </head>
131 <body>
132     <div class="card">
133         
134         <h2>Varshitha</h2>
135         <p>CSE Student</p>
136         <div class="social-links">
137             <a href="#" target="_blank" class="linkedin">LinkedIn</a>
138             <a href="#" target="_blank" class="twitter">Twitter</a>
139             <a href="#" target="_blank" class="github">GitHub</a>
140         </div>
141     </div>
142 </body>
143 </html>

```



Task 3 -Develop a dynamic shopping cart system.

Requirements:• Add, remove, and update product quantities.• Display real-time total price.• Use AI to:

o Manage cart state using JavaScript objects.

```
<!DOCTYPE html>
<html>
<head>
  <title>Shopping Cart</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f0f0;
      margin: 0;
      padding: 20px;
    }
    .cart {
      max-width: 600px;
      margin: auto;
      background-color: white;
      padding: 20px;
      border-radius: 5px;
      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
    }
    .cart h2 {
      margin-top: 0;
    }
    .cart-item {
      display: flex;
      justify-content: space-between;
      align-items: center;
      margin-bottom: 15px;
    }
    .cart-item input {
      width: 50px;
    }
    .total {
      font-size: 18px;
      font-weight: bold;
    }
  </style>
</head>
</html>
```

```

</style>
</head>
<body>
  <div class="cart">
    <h2>Shopping Cart</h2>
    <div id="cart-items"></div>
    <p class="total">Total: $<span id="total-price">0.00</span></p>
  </div>

  <script>
    const cart = {
      items: [],
      addItem(name, price) {
        const existingItem = this.items.find(item => item.name === name);
        if (existingItem) {
          existingItem.quantity += 1;
        } else {
          this.items.push({ name, price, quantity: 1 });
        }
        this.updateCart();
      },
      removeItem(name) {
        this.items = this.items.filter(item => item.name !== name);
        this.updateCart();
      },
      updateQuantity(name, quantity) {
        const item = this.items.find(item => item.name === name);
        if (item) {
          item.quantity = quantity;
          if (item.quantity <= 0) {
            this.removeItem(name);
          } else {
            this.updateCart();
          }
        }
      }
    };
  </script>

```

```

    calculateTotal() {
      return this.items.reduce((total, item) => total + item.price * item.quantity, 0).toFixed(2);
    },
    updateCart() {
      const cartItemsContainer = document.getElementById('cart-items');
      cartItemsContainer.innerHTML = '';
      this.items.forEach(item => {
        const cartItem = document.createElement('div');
        cartItem.className = 'cart-item';
        cartItem.innerHTML = `
          <span>${item.name} - ${item.price}</span>
          <input type="number" value="${item.quantity}" min="1" onchange="cart.updateQuantity('${item.name}', parseInt(this.value))" />
          <button onclick="cart.removeItem('${item.name}')">Remove</button>
        `;
        cartItemsContainer.appendChild(cartItem);
      });
      document.getElementById('total-price').textContent = this.calculateTotal();
    }
  };

  // Example products
  cart.addItem('Product 1', 19.99);
  cart.addItem('Product 2', 9.99);
</script>
</body>
</html>

```

Shopping Cart

Product 1 - \$19.99	2	<button>Remove</button>
Product 2 - \$9.99	3	<button>Remove</button>
Total: \$69.95		

Task 4 – Role-Based Access Control Application

Create a role-based web application.

Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
 - o Design access control logic.
 - o Maintain secure state transitions.
 - o Create scalable UI components.

```

2 <!DOCTYPE html>
3 <html>
4 <head>
5 <title>Role-Based Web Application</title>
6 <style>
7     body {
8         font-family: Arial, sans-serif;
9         background-color: #f0f0f0;
0         margin: 0;
1         padding: 20px;
2     }
3     .container {
4         max-width: 600px;
5         margin: auto;
6         background-color: white;
7         padding: 20px;
8         border-radius: 5px;
9         box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
0     }
1     .hidden {
2         display: none;
3     }
4     .admin, .editor, .user {
5         padding: 20px 40px;
6         border-radius: 10px;
7     }
8     .admin {
9         background-color: #ff4d4d;
0         color: white;
1     }
2     .editor {
3         background-color: #4d79ff;
4         color: white;
5     }

```

```

}
.admin:hover {
|   background-color: #e60000;
}
.editor:hover {
|   background-color: #3366ff;
}
.user:hover {
|   background-color: #45a049;
}
.admin, .editor, .user {
|   margin: 10px 0;
|   cursor: pointer;
}
/* Add hover effects for better user experience */
.admin:hover, .editor:hover, .user:hover {
|   transform: scale(1.05);
|   box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
}
.admin, .editor, .user {
|   transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.role-buttons {
|   display: flex;
|   gap: 10px;
|   margin-bottom: 20px;
|   justify-content: center;
}
.role-buttons button {
|   padding: 10px 20px;
|   border: none;
|   border-radius: 5px;
|   cursor: pointer;
|   font-size: 16px;
|   transition: background-color 0.3s;
}
.section {
|   padding: 20px;
|   border-radius: 10px;
|   margin-bottom: 10px;
|   color: white;
}
section h2 {

```

```

    }
    .form-group {
      margin-bottom: 15px;
    }
    .form-group label {
      display: block;
      margin-bottom: 5px;
    }
    .form-group input, .form-group textarea, .form-group button {
      width: 100%;
      padding: 8px;
      border: none;
      border-radius: 5px;
    }
    .form-group button {
      background-color: #333;
      color: white;
      cursor: pointer;
      margin-top: 10px;
    }
    .form-group button:hover {
      background-color: #555;
    }
    .content-display {
      background-color: #f9f9f9;
      padding: 10px;
      border-radius: 5px;
      margin-bottom: 10px;
    }
  }
</style>
</head>
<body>
  <div class="container">
    <h1>Role-Based Web Application</h1>
    <p>Current Role: <span id="current-role">Editor</span></p>
    <div class="role-buttons">
      <button onclick="changeRole('admin')" class="admin">Switch to Admin</button>
      <button onclick="changeRole('editor')" class="editor">Switch to Editor</button>
      <button onclick="changeRole('user')" class="user">Switch to User</button>
    </div>
    <div id="admin-section" class="section admin hidden">
      <h3>Admin Section: Manage Users, View Reports</h3>
      <div class="form-group">
        <label for="new-user">Add New User:</label>
        <input type="text" id="new-user" placeholder="Enter username">
        <button onclick="addUser()">Add User</button>
      </div>
      <div id="user-list"></div>
    </div>
    <div id="editor-section" class="section editor hidden">
      <h3>Editor Section: Edit Content, Publish Articles</h3>
      <div class="form-group">

```


Role-Based Web Application

Current Role: Admin

Switch to Admin

Switch to Editor

Switch to User

Admin Section: Manage Users, View Reports

Add New User:

Enter username

Add User

User List:

- Alice
- Bob
- jii
- Varadha
- Salaar

Role-Based Web Application

Current Role: Editor

Switch to Admin

Switch to Editor

Switch to User

Editor Section: Edit Content, Publish Articles

Edit Content:

Content is edited

Publish

Role-Based Web Application

Current Role: User

Switch to Admin

Switch to Editor

Switch to User

User Section: View Content, Comment

Content is edited

Add Comment:

Your comment...

Comment

Comments:

Hello,

Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

- Switch languages dynamically.
- Persist language preference.
- Use AI to:
 - o Structure translation files.
 - o Handle RTL/LTR layouts.
 - o Improve accessibility.

```

73 <!DOCTYPE html>
74 <html>
75 <head>
76   <title>Multilingual Website</title>
77   <style>
78     body {
79       font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
80       background: linear-gradient(135deg, #1e3c72 0%, #2a5298 100%);
81       color: #222;
82       margin: 0;
83       padding: 30px;
84       min-height: 100vh;
85     }
86     .wrapper {
87       max-width: 1100px;
88       margin: 0 auto;
89       padding: 20px;
90     }
91     .hero {
92       background: rgba(255,255,255,0.12);
93       border: 1px solid rgba(255,255,255,0.2);
94       border-radius: 14px;
95       padding: 30px;
96       margin-bottom: 20px;
97       color: #fff;
98       box-shadow: 0 12px 40px rgba(0,0,0,0.25);
99       backdrop-filter: blur(5px);
100    }
101    .hero h1 {
102      margin-top: 0;
103      font-size: 32px;
104      letter-spacing: 1px;
105    }
106    .hero p {
107      font-size: 18px;
108      opacity: 0.95;
109    }
110    .language-selector {
111      margin-bottom: 20px;
112    }
113    .language-selector label {
114      font-weight: bold;
115      color: #fff;
116      margin-right: 10px;
117    }
118    .language-selector select {
119      padding: 10px;
120      border-radius: 8px;
121      border: none;
122      font-size: 16px;
123      box-shadow: 0 4px 10px rgba(0,0,0,0.2);
124    }
125    .language-selector select:hover {
126      background-color: #eaeaea;

```

```

6   .hero p {
7       font-size: 18px;
8       opacity: 0.95;
9   }
10  .language-selector {
11      margin-bottom: 20px;
12  }
13  .language-selector label {
14      font-weight: bold;
15      color: #fff;
16      margin-right: 10px;
17  }
18  .language-selector select {
19      padding: 10px;
20      border-radius: 8px;
21      border: none;
22      font-size: 16px;
23      box-shadow: 0 4px 10px #00000020;
24  }
25  .language-selector select:hover {
26      background-color: #eaeaea;
27  }
28  .language-selector select:focus {
29      outline: none;
30      box-shadow: 0 0 8px #00000020;
31  }
32  .content {
33      background-color: #00000020;
34      padding: 25px;
35      border-radius: 12px;
36      box-shadow: 0 10px 30px #00000015;
37      transition: all 0.3s ease;
38  }
39  .content:hover {
40      transform: translateY(-3px);
41      box-shadow: 0 12px 36px #00000025;
42  }
43  .content h1 {
44      color: #0f548a;
45      margin-top: 0;
46  }
47  .content p {
48      color: #333;
49      font-size: 1rem;
50      line-height: 1.6;
51  }
52  .content ul {
53      padding-left: 20px;
54  }
55  .content ul li {
56      margin-bottom: 8px;
57  }

```

```

91 <body>
92   <div class="wrapper">
93     <div class="hero" role="banner">
94       <h1>Multilingual Website Demo</h1>
95       <p>Switch content between English, Spanish, and Arabic with full RTL/LTR support. Preference is saved in localStorage for ne
96     </div>
97     <section class="language-selector" aria-labelledby="language-label">
98       <label id="language-label" for="language">Select Language:</label>
99       <select id="language" onchange="changeLanguage(this.value)">
100         <option value="en">English</option>
101         <option value="es">Spanish</option>
102         <option value="ar">Arabic</option>
103       </select>
104     </section>
105     <section class="content ltr" id="content" aria-live="polite" aria-atomic="true">
106       <h1 id="main-title">AI Insights in Multilingual Mode</h1>
107       <p id="main-description">Explore how AI can power content translation, personalization, and accessibility on your website.</p>
108       <ul id="feature-list">
109         <li id="feature-1">AI-powered content translation</li>
110         <li id="feature-2">Language-aware layout adaptation (RTL/LTR)</li>
111         <li id="feature-3">Persistent language preference via localStorage</li>
112         <li id="feature-4">Interactive cards with AI concept summaries</li>
113       </ul>
114     </section>
115     <div class="translation-cards">
116       <article class="card" aria-label="AI model overview">
117         <h3 id="card1-title">AI Model Basics</h3>
118         <p id="card1-text">AI systems learn patterns from data to make predictions and generate content automatically.</p>
119       </article>
120       <article class="card" aria-label="NLP overview">
121         <h3 id="card2-title">Natural Language Processing</h3>
122         <p id="card2-text">NLP is the AI subfield that enables machines to understand and generate human language.</p>
123       </article>
124       <article class="card" aria-label="ML overview">
125         <h3 id="card3-title">Machine Learning</h3>
126         <p id="card3-text">ML algorithms improve over time by learning from examples, powering features like recommendation syst
127       </article>
128     </div>
129   </div>
130   <script>
131     const translations = {
132       en: {
133         heroTitle: "AI & Multilingual Web Experience",
134         heroDesc: "Learn how AI can make content accessible globally by automatically adapting language and layout.",
135         selectorLabel: "Select Language:",
136         mainTitle: "AI Insights in Multilingual Mode",
137         mainDesc: "Explore how AI can power content translation, personalization, and accessibility on your website.",
138         feature1: "AI-powered content translation",
139         feature2: "Language-aware layout adaptation (RTL/LTR)",
140         feature3: "Persistent language preference via localStorage",
141         feature4: "Interactive cards with AI concept summaries",
142         card1Title: "AI Model Basics",
143         card1Text: "AI systems learn patterns from data to make predictions and generate content automatically.",

```

```

32     }
33   };
34   function changeLanguage(lang) {
35     const translation = translations[lang];
36     document.querySelector('.hero h1').textContent = translation.heroTitle;
37     document.querySelector('.hero p').textContent = translation.heroDesc;
38     document.querySelector('label[for="language"]').textContent = translation.selectorLabel;
39     document.getElementById('main-title').textContent = translation.mainTitle;
40     document.getElementById('main-description').textContent = translation.mainDesc;
41     document.getElementById('feature-1').textContent = translation.feature1;
42     document.getElementById('feature-2').textContent = translation.feature2;
43     document.getElementById('feature-3').textContent = translation.feature3;
44     document.getElementById('feature-4').textContent = translation.feature4;
45     document.getElementById('card1-title').textContent = translation.card1Title;
46     document.getElementById('card1-text').textContent = translation.card1Text;
47     document.getElementById('card2-title').textContent = translation.card2Title;
48     document.getElementById('card2-text').textContent = translation.card2Text;
49     document.getElementById('card3-title').textContent = translation.card3Title;
50     document.getElementById('card3-text').textContent = translation.card3Text;
51     const content = document.getElementById('content');
52     if (lang === 'ar') {
53       content.classList.remove('ltr');
54       content.classList.add('rtl');
55       document.body.style.direction = 'rtl';
56     } else {
57       content.classList.remove('rtl');
58       content.classList.add('ltr');
59       document.body.style.direction = 'ltr';
60     }
61     localStorage.setItem('preferredLanguage', lang);
62   }
63   window.onload = function() {
64     const preferredLanguage = localStorage.getItem('preferredLanguage') || 'en';
65     document.getElementById('language').value = preferredLanguage;
66     changeLanguage(preferredLanguage);
67   };
68   </script>
69 </body></html>

```

الذكاء الاصطناعي وتجربة الويب متعددة اللغات

تعرف على كيفية تمكين الذكاء الاصطناعي المحتوى ليكون متاحًا عالميًا من خلال تكييف اللغة وتصميم الصفحة.

اختر اللغة: Arabic

رؤى الذكاء الاصطناعي في وضع متعدد اللغات

اكتشف كيف يمكن للذكاء الاصطناعي تعزيز ترجمة المحتوى والتخصيص وإمكانية الوصول.

- ترجمة المحتوى بواسطة الذكاء الاصطناعي
- تكيف اتجاه التخطيط RTL/LTR حسب اللغة
- حفظ تفضيل اللغة باستخدام localStorage
- بطاقات تفاعلية مع ملخصات مفاهيم الذكاء الاصطناعي

التعلم الآلي

تتحسن خوارزميات التعلم الآلي بمرور الوقت عبر الأمثلة وتفاعل المستخدمين.

معالجة اللغة الطبيعية

تمكن معالجة اللغة الطبيعية الآلات من فهم وإنتاج اللغة البشرية.

أساسيات الذكاء الاصطناعي

تتعلم أنظمة الذكاء الاصطناعي الأنماط من البيانات لتوليد المحتوى والتنبؤ.



Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:
 - o Manage exam state.
 - o Improve exam UX.
 - o Ensure accessibility.


```

<!DOCTYPE html>
<html>
<head>
  <title>Online Exam Interface</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f0f0;
      margin: 0;
      padding: 20px;
    }
    .exam-container {
      max-width: 600px;
      margin: auto;
      background-color: white;
      padding: 20px;
      border-radius: 5px;
      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
    }
    .timer {
      font-size: 18px;
      font-weight: bold;
      margin-bottom: 20px;
    }
    .question {
      margin-bottom: 15px;
    }
    .submit-btn {
      padding: 10px 20px;
      background-color: #333;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }
    .submit-btn:hover {
      background-color: #555;
    }
  </style>
</head>
<body>
  <div class="exam-container">
    <div class="timer" id="timer">Time Remaining: 10:00</div>
    <div class="question">
      <p>1. What is the capital of France?</p>
      <input type="text" id="q1" placeholder="Your answer">
    </div>
    <div class="question">
      <p>2. What is 2 + 2?</p>
      <input type="text" id="q2" placeholder="Your answer">
    </div>
    <div class="question">

```

```

2         return;
3     }
4     timeRemaining -= 1;
5     timerElement.textContent = `Time Remaining: ${formatTime(timeRemaining)}`;
6 }
7
8 function submitExam(autoSubmit = false) {
9     if (examSubmitted) {
10         alert('Exam already submitted.');
```

```

11         return;
12     }
13
14     examSubmitted = true;
15     submitButton.disabled = true;
16     submitButton.textContent = 'Submitted';
17     clearInterval(timerInterval);
18
19     const q1 = document.getElementById('q1').value.trim();
20     const q2 = document.getElementById('q2').value.trim();
21     const q3 = document.getElementById('q3').value.trim();
22     const q4 = document.getElementById('q4').value.trim();
23
24     const assessment = [];
25     assessment.push(`Q1: ${q1} || 'No answer'`);
26     assessment.push(`Q2: ${q2} || 'No answer'`);
27     assessment.push(`Q3: ${q3} || 'No answer'`);
28     assessment.push(`Q4: ${q4} || 'No answer'`);
29
30     if (autoSubmit) {
31         alert('Time is up! Your exam has been submitted automatically.');
```

```

32     } else {
33         alert('Exam submitted successfully.');
```

```

34     }
35
36     const resultSummary = document.createElement('div');
37     resultSummary.style.marginTop = '20px';
38     resultSummary.style.padding = '10px';
39     resultSummary.style.backgroundColor = '#e8f4ff';
40     resultSummary.style.border = '1px solid #007bff';
41     resultSummary.innerHTML = `<strong>Submitted Answers:</strong><br>${assessment.join('<br>')}`;
42     document.querySelector('.exam-container').appendChild(resultSummary);
43 }
44
45 const timerInterval = setInterval(updateTimer, 1000);
46 timerElement.textContent = `Time Remaining: ${formatTime(timeRemaining)}`;
47 </script>
48 </body></html>

```

Time Remaining: 09:01

1. What is the capital of France?

2. What is 2 + 2?

3. Who is considered the father of modern AI?

4. What does "NLP" stand for?